

My Contribution — Source Code Handling, Lexer and Parser

Project: BanglaLang Compiler

Language: C++17

1. Source Code Handling

I worked on the source code handling part of the compiler.

My work included:

- Reading the source program from an input file.
- Passing the source code to the lexical analyzer.
- Managing the source code as input for the compilation process.
- Handling source file opening and reading errors.

2. Lexical Analysis (Lexer)

I implemented the Lexer for the BanglaLang compiler.

The Lexer performs the following tasks:

- Converts source code into tokens.
- Identifies keywords such as:
 - সংখ্যা
 - দশমিক
 - যদি
 - নاهলে
 - যতক্ষণ
 - দেখাও
 - শুরু
 - শেষ
- Recognizes identifiers.
- Recognizes integer and decimal literals.
- Recognizes arithmetic operators such as:

- +
- -
- *
- /
- %
- Recognizes comparison operators such as:
 - ==
 - !=
 - <
 - <=
 - >
 - >=
- Recognizes assignment operator =.
- Handles parentheses, braces, commas and semicolons.
- Skips whitespace and single-line comments.
- Tracks line and column positions.
- Detects invalid or unsupported characters.
- Generates an End-of-File token at the end of the source program.

3. Syntax Analysis (Parser)

I also implemented the Parser part of the compiler.

The Parser takes the tokens generated by the Lexer and checks the structure of the program.

My Parser handles:

- Program structure.
- Variable declarations.
- Variable assignments.
- Print statements.
- if-else statements.
- while loops.

- Arithmetic expressions.
- Comparison expressions.
- Parenthesized expressions.
- Unary expressions.
- Operator precedence.
- Syntax error detection.
- Error synchronization and recovery.

The Parser also builds the required Abstract Syntax Tree (AST) structures for expressions and statements.

4. AST Structures

The AST structures used in my Parser include:

Expression Nodes

- LiteralExpr
- VariableExpr
- UnaryExpr
- BinaryExpr

Statement Nodes

- VarDeclStmt
- AssignStmt
- PrintStmt
- IfStmt
- WhileStmt

5. Technologies Used

- C++
- C++17
- File Handling
- Lexical Analysis
- Syntax Analysis

- Abstract Syntax Tree (AST)

6. Contribution Scope

My contribution to the BanglaLang Compiler project focused on:

1. Source Code Handling
2. Lexical Analysis (Lexer)
3. Syntax Analysis (Parser)
4. Required AST structures used by the Parser

The Semantic Analysis and Code Generation parts were developed separately from my contribution.